

1. Define System and Its Types

Definition of System

A **system** is a group of **interrelated and interconnected elements** that work together to achieve a **common objective**.

Simple definition:

A system is a group of parts that work together to achieve a specific goal.

Example: A computer system consists of CPU, memory, keyboard, mouse, and other components that work together to process information.

Types of Systems

1. Open System

An **open system** interacts with its external environment. It receives **inputs** from the environment and produces **outputs**.

Example: A business organization receives raw materials, money, and employees from outside and produces goods or services.

2. Closed System

A **closed system** has little or no interaction with its external environment.

Example: A simple mechanical clock can be considered a relatively closed system.

3. Physical System

A system made up of **physical/tangible objects**.

Example: Computer system, transportation system, manufacturing system.

4. Abstract System

A system made up of **ideas, concepts, or theories** rather than physical objects.

Example: Mathematical system, accounting system.

5. Natural System

A system that occurs **naturally without being created by humans**.

Example: Solar system, ecosystem.

6. Man-Made System

A system **designed and created by humans** to achieve a particular purpose.

Example: Banking system, education system, production system.

2. Difference Between Product Management and Industrial Management

Product Management

Product management is the process of **planning, developing, marketing, and managing a product throughout its life cycle.**

It mainly focuses on **the product and customer needs.**

Industrial Management

Industrial management is the process of **planning, organizing, directing, and controlling industrial activities and resources** to achieve organizational goals efficiently.

It focuses on the **overall management of an industrial organization.**

Difference

Product Management

Focuses mainly on a **specific product.**

Concerned with **product planning and development.**

Focuses strongly on **customer needs and market demand.**

Manages the **product life cycle.**

Includes product design, pricing, marketing, and improvement.

Main goal is to make the **product successful in the market.**

Easy way to remember:

Product Management → Product + Customer + Market

Industrial Management → Factory/Industry + People + Resources + Production

Industrial Management

Focuses on the **overall industrial organization.**

Concerned with **overall planning and management.**

Focuses on **production, workers, resources, finance, etc.**

Manages **industrial operations and resources.**

Includes planning, organizing, staffing, directing, and controlling.

Main goal is to achieve **efficient industrial operations and organizational goals.**